Probability theory

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Outline

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- Marginal, conditional and joint probability of discrete and continuous variables
- Probability density and probability distribution
- Marginal, conditional and joint probability of continuous variables
- Statistical and conditional independence
- Mean, variance and covariance
- Correlation
- Bayes' rule
- Principal probability distributions
- Information Theory

Probability of discrete variables

- A discrete causal variable X assumes values in the discrete set $\mathcal{X} = \{x_i, i = 1,2, \dots\}$
- Over N observations of X, c_i times $X = x_i$

$$p(X = x_i) = \frac{c_i}{N},$$

$$\forall\, x_i, 0 \leq p(X = x_i) \leq 1$$

$$\sum_{x_i} p(X = x_i) = 1$$

Probability of discrete variables

- Function expressing that the probability $P(x) = p\ (X = x)$, $x \in \mathcal{X}$ for a certain discrete random variable $\mathcal{X}$, is called discrete probability distribution

- The discrete probability distribution can be described by a Probability Mass Function that assigns a probability to each value in the image of $\mathcal{X}$.

Marginal probability

Given a known joint distribution of two discrete random variables, X and Y, the marginal distribution of X is the probability distribution of X when the values of Y are not taken into consideration. This can be calculated by summing the joint probability distribution over all values of Y.

$$p(X = x_i) = \sum_j p(X = x_i, Y = y_j)$$

sum rule

Marginal probability

	x_1	x_2	x_3	x_4	$p_y(Y) \downarrow$
y_1	$\frac{1}{8}$	$\frac{1}{16}$	$\frac{1}{32}$	$\frac{1}{32}$	$\frac{1}{4}$
y_2	$\frac{1}{16}$	$\frac{1}{8}$	$\frac{1}{32}$	$\frac{1}{32}$	$\frac{1}{4}$
y_3	$\frac{1}{16}$	$\frac{1}{16}$	$\frac{1}{16}$	$\frac{1}{16}$	$\frac{1}{4}$
y_4	$\frac{1}{4}$	0	0	0	$\frac{1}{4}$
$p_x(X) \rightarrow$	$\frac{1}{2}$	$\frac{1}{4}$	$\frac{1}{8}$	$\frac{1}{8}$	1

Conditional probability

Probability that $Y = y_j$ if $X = x_i$

$$p(Y = y_j | X = x_i) = \frac{P(X = x_i, Y = y_i)}{P(X = x_i)} = \frac{n_{ij}}{c_i}$$

n_{ij} number of observation where $Y = y_j$ if $X = x_i$

c_i number of observation there $X = x_i$

Joint probability

Probability that $Y = y_j$ *and* $X = x_i$

$$p(X = x_i, Y = y_i) = \frac{n_{ij}}{N} = \frac{n_{ij}}{c_i} * \frac{c_i}{N} = p\left(Y = y_j \middle| X = x_i\right) * p(X = x_i)$$

product rule

n_{ij} number of observation where $Y = y_j$ if $X = x_i$

c_i number of observation there $X = x_i$

Continuous variables

If X is a continuous variable in $\mathbb{R}$

$$p(x \in [a,b]) = \int_a^b p(x)dx$$

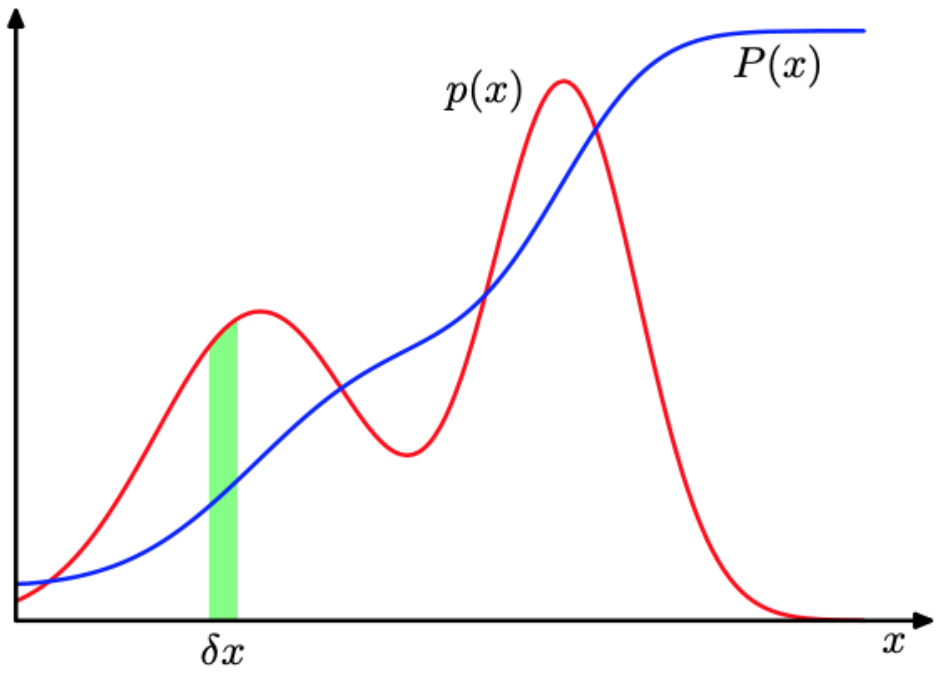
Probability density and probability distribution

Probability Distribution Function

expresses the probability *P(x)=p(X=x)* given X continuous random variable

$$p(x) \geq 0$$

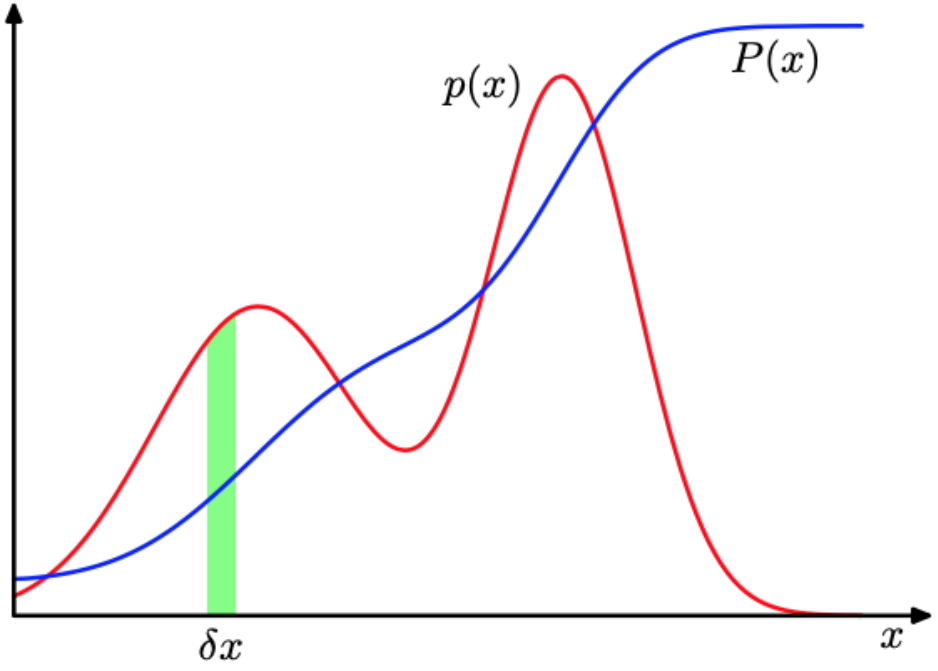
$$\int_{-\infty}^{+\infty} p(x)dx = 1$$



Probability density and probability distribution

Cumulative distribution function

$$P(z) = \int_{-\infty}^{z} p(x)dx \equiv p(x \leq z)$$



Marginal, conditional and joint probability of continuous variables

Marginal probability

$$p(x) = \int p(x, y)dy$$

Conditional probability

$$P(Y = y|X = x) = \frac{P(Y = y, X = x)}{P(X = x)}$$

Joint probability

$$P\left(x^{(1)}, \dots, x^{n}\right) = P(x^{1}) \prod_{i=2}^{n} P(x^{(i)}|x^{(1)}, \dots, x^{(i-1)})$$

Statistical and conditional independence

Statistical indipendence

$$\forall x \in X, y \in Y, \qquad p(X = x, Y = y) = p(X = x)p(Y = y)$$

Conditional indipendence

$$\forall x \in X, y \in Y, z \in Z,$$
$$p(X = x, Y = y | Z = z) = p(X = x | Z = z)p(Y = y | Z = z)$$

Mean, variance and covariance

Mean or expected value is a linear operator and the output is a number and not more a random variable

$$\mathbb{E}_{x\sim P}[f(x)] = \sum_{x} P(x)f(x)$$

$$\mathbb{E}_{x\sim P}[f(x)] = \int P(x)f(x)dx$$

$$\mathbb{E}_{x}[\alpha f(x) + \beta g(x)] = \alpha\mathbb{E}_{x}[f(x)] + \beta\mathbb{E}_{x}[g(x)]$$

Mean, variance and covariance

Variance

It is the expected value of the difference between f(x) and the square of its expected value

$$\mathrm{Var}\big(f(x)\big) = \mathbb{E}[(f(x) - \mathbb{E}[f(x)])^2] =$$
$$= \mathbb{E}[f(x)^2] - \mathbb{E}[f(x)]^2$$

Mean, variance and covariance

Variance's Property, under statistical independence hypothesis

$$\begin{aligned}\mathrm{Var}(\alpha \cdot f(x)) =& \mathbb{E}\left[\alpha^2 \cdot f(x)^2\right] - \boxed{\mathbb{E}\left[\alpha \cdot f(x)\right]^2} = \\ & \alpha^2 \cdot \mathbb{E}\left[f(x)^2\right] - \alpha^2 \cdot \mathbb{E}\left[f(x)\right]^2 = \\ & \alpha^2 \cdot \left(\mathbb{E}\left[f(x)^2\right] - \mathbb{E}\left[f(x)\right]^2\right) \equiv \alpha^2 \cdot \mathrm{Var}(f(x))\end{aligned}$$

$$(E[\alpha X])^2 = (\alpha E[X])^2 = \alpha^2 (E[X])^2$$

Mean, variance and covariance

Variance's Property, under statistical independence hypothesis

$$\begin{aligned}\mathrm{Var}(f(x)+g(y)) =& \mathbb{E}\left[(f(x)+g(y))^2\right] - \mathbb{E}\left[(f(x)+g(y))\right]^2 = \\ &= \mathrm{Var}(f(x)) + \mathrm{Var}(g(y))\end{aligned}$$

Variance is usually referred to as σ^2, while σ is the <u>standard deviation</u>

Mean, variance and covariance

Covariance

Expresses the linear relationship of two random variables

$$\mathrm{Cov}(f(x), g(y)) = \mathbb{E}[(f(x) - \mathbb{E}[f(x)])(g(y) - \mathbb{E}[g(y)])] =$$
$$= \mathbb{E}[f(x)g(y)] - \mathbb{E}[f(x)]\mathbb{E}[g(y)]$$

Mean, variance and covariance

A covariance matrix can be calculated between two columns vectors $\mathbf{x}, \mathbf{y} \in \mathbb{R}^n$, each element is a random variable

$$\begin{aligned}\mathrm{Cov}(\mathbf{x}, \mathbf{y}) &= \mathbb{E}\left(\{\mathbf{x} - \mathbb{E}[\mathbf{x}]\}\left\{\mathbf{y}^{\mathrm{T}} - \mathbb{E}\left[\mathbf{y}^{\mathrm{T}}\right]\right\}\right) \\ &= \mathbb{E}\left[\mathbf{x}\mathbf{y}^{\mathrm{T}}\right] - \mathbb{E}[\mathbf{x}]\mathbb{E}\left[\mathbf{y}^{\mathrm{T}}\right] \in \mathbb{R}^{\mathrm{n}\times\mathrm{n}}\end{aligned}$$

Covariance in **x** elements

$$\begin{aligned}\mathrm{Cov}(\mathbf{x}, \mathbf{x}) &= \mathbb{E}\left(\{\mathbf{x} - \mathbb{E}[\mathbf{x}]\}\left\{\mathbf{x}^{\mathrm{T}} - \mathbb{E}\left[\mathbf{x}^{\mathrm{T}}\right]\right\}\right) \\ &= \mathbb{E}\left[\mathbf{x}^2\right] - \mathbb{E}[\mathbf{x}]\mathbb{E}\left[\mathbf{x}^{\mathrm{T}}\right] = \mathbb{E}\left[\mathbf{x}^2\right] - \mathbb{E}\left[\mathbf{x}\right]^2 \equiv \mathrm{Var}(\mathbf{x})\end{aligned}$$

Mean, variance and covariance

Covariance matrix of a data set *D with n* record $\mathbf{x}_k = \{x_k^1, \ldots, x_k^d\}$, $k = 1, \ldots, n$

$$C = \frac{D^T D}{n} - \bar{\mu}^T \bar{\mu}, \;\; \bar{\mu} = \left\{ \mathbb{E}\left[x_k^1\right], \ldots, \mathbb{E}\left[x_k^d\right] \right\}$$

Its element can be defined as

$$c_{ij} = \frac{\sum_{k=1}^{n} x_k^i x_k^j}{n} - \mu_i \mu_j \quad \forall i, j \in \{1 \ldots d\}$$

$$c_{ii} = \frac{\sum_{k=1}^{n} {x_k^i}^2}{n} - \mu_i^2 \equiv \sigma_i^2$$

Correlation

- Covariance matrix over a data set *D*, expresses how different components in $\mathbf{x}_k$ (feature) are linearly realted to each other.
- It is a positive semi-definite matrix
- Another index can be derived from covariance, it is used to indicate how related two features are and it is the correlation coefficient or Pearson correlation index.

Correlation

Let *X* and *Y* be two random variables

$$\Sigma_{X,Y} = \begin{bmatrix} \sigma_X^2 & \mathrm{Cov}(X,Y) \\ \mathrm{Cov}(X,Y) & \sigma_Y^2 \end{bmatrix}$$

$$\det \Sigma_{X,Y} = \sigma_X^2 \sigma_Y^2 - \mathrm{Cov}(X,Y)^2 \geq 0$$

Correlation coefficient is defined as

$$\rho_{X,Y} = \frac{\mathrm{Cov}(X,Y)}{\sigma_X \sigma_Y} \in [-1, 1]$$

Correlation

$$\rho_{jk} = \frac{\mathrm{Cov}\left(x^{(j)}, x^{(k)}\right)}{\sigma_j \cdot \sigma_k}$$

$$\rho_{jk} = \frac{\mathbb{E}\left[\left(x^{(j)} - \mathbb{E}\left[x^{(j)}\right]\right)\left(x^{(k)} - \mathbb{E}\left[x^{(k)}\right]\right)\right]}{\sqrt{\mathbb{E}\left[\left(x^{(j)} - \mathbb{E}\left[x^{(j)}\right]\right)^2\right]} \cdot \sqrt{\mathbb{E}\left[\left(x^{(k)} - \mathbb{E}\left[x^{(k)}\right]\right)^2\right]}}$$

Bayes' rule

It is possibile to express the joint probability *P(x,y)* of two correlated variables, from their conditional probability

$$P(\mathrm{x}|\mathrm{y}) = \frac{P(\mathrm{y}|\mathrm{x})P(\mathrm{x})}{P(\mathrm{y})}$$

Bayes' rule

Given data set $\mathcal{D}$, we want to learn a model described by a vector of parameter $\mathbf{w}$

We will obtain the best model when the posterior will be maximum $P(\mathbf{w} \mid \mathcal{D})$

what we can estimate *after* data observation

Bayes' rule

Bayes' rule correlates posterior with

the prior P(**w**) (apriori knowledge) over the model

the likelihood of the model, expressed by the probability to predict $\boldsymbol{\mathcal{D}}$ given model's parameters **w**, that is P($\boldsymbol{\mathcal{D}}$ | **w**)

$$P(\mathbf{w}|\boldsymbol{\mathcal{D}}) = \frac{P(\boldsymbol{\mathcal{D}}|\mathbf{w})P(\mathbf{w})}{P(\boldsymbol{\mathcal{D}})}$$

$$\text{posterior} \propto \text{likelihood} \times \text{prior}$$

Bayes' rule

A priori knowledge P($\boldsymbol{\mathcal{D}}$) can be marginalized and expressed as

$$P(\boldsymbol{\mathcal{D}}) = \int P(\boldsymbol{\mathcal{D}}|\mathbf{w})P(\mathbf{w})d\mathbf{w}$$

Generally, it can be seen as a costant since it is the evidence of the data set despite the parameters

Principal probability distributions

We will deal with diverse type of data

- Multidimensional
- Categorical
- Discrete Series
- …

Different types of probability distributions can model data

We need to know them because *Machine Learning is the process of estimating these probability distributions from the data that the model observes*

Principal probability distributions

- Bernoulli distribution for binary variables

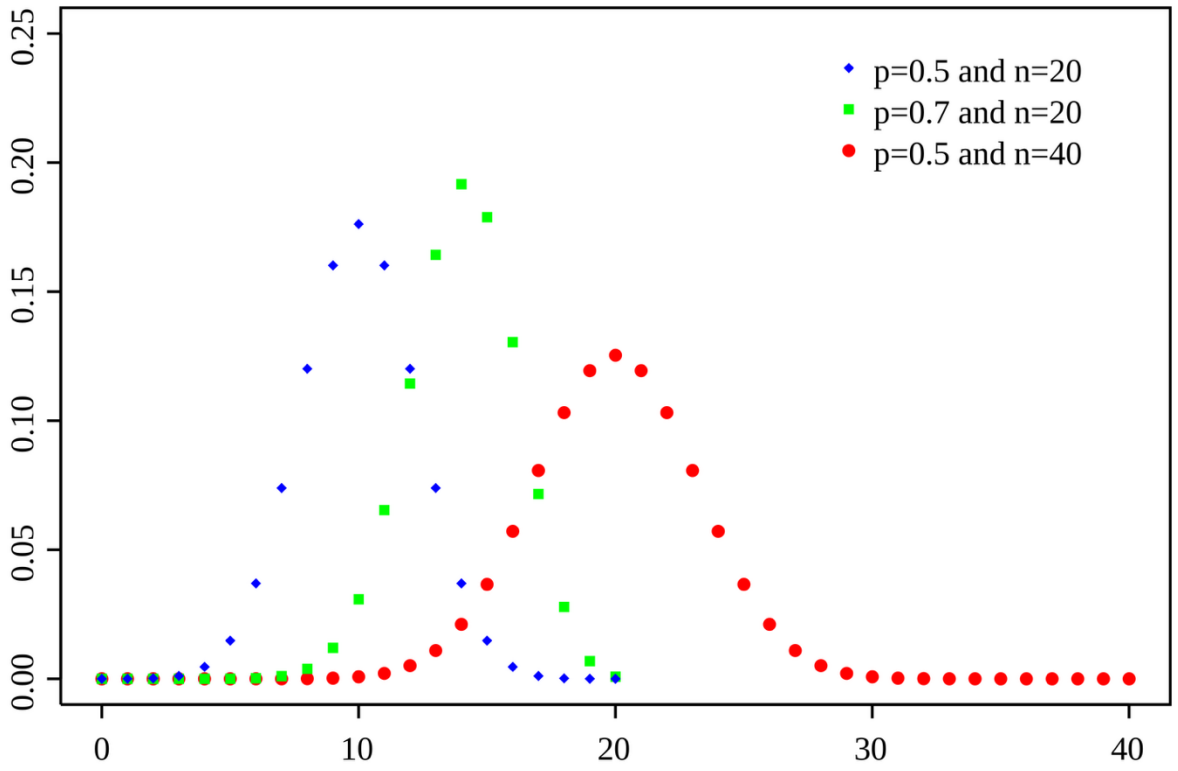
$$P(\mathrm{x} = 1) = \phi$$
$$P(\mathrm{x} = 0) = 1 - \phi$$
$$P(\mathrm{x} = x) = \phi^{x}(1 - \phi)^{1-x}$$
$$\mathbb{E}_{\mathrm{x}}[\mathrm{x}] = \phi$$
$$\mathrm{Var}_{\mathrm{x}}(\mathrm{x}) = \phi(1 - \phi)$$

Principal probability distributions

- Bernoulli distribution for binary variables

$$\mathrm{Bin}\,(m \mid N, \phi) = \binom{N}{m} \phi^m (1-\phi)^{N-m}$$

m observation of x=1 over N attempts



Principal probability distributions

- Multinoulli distribution (categorical distribution) in which x is a variable that may assume one among *K* different states

$$\mathrm{x} = (0_1, 0_2, \dots, 1_k, \dots, O_K)^T$$

$$p(\mathrm{x}|\boldsymbol{\mu}) = \prod_{k=1}^{K} \mu_k^{x_k}$$

$$\boldsymbol{\mu} = (\mu_1, \dots \mu_K)^T$$

$$\mu_k = p(x_k = 1)$$

Principal probability distributions

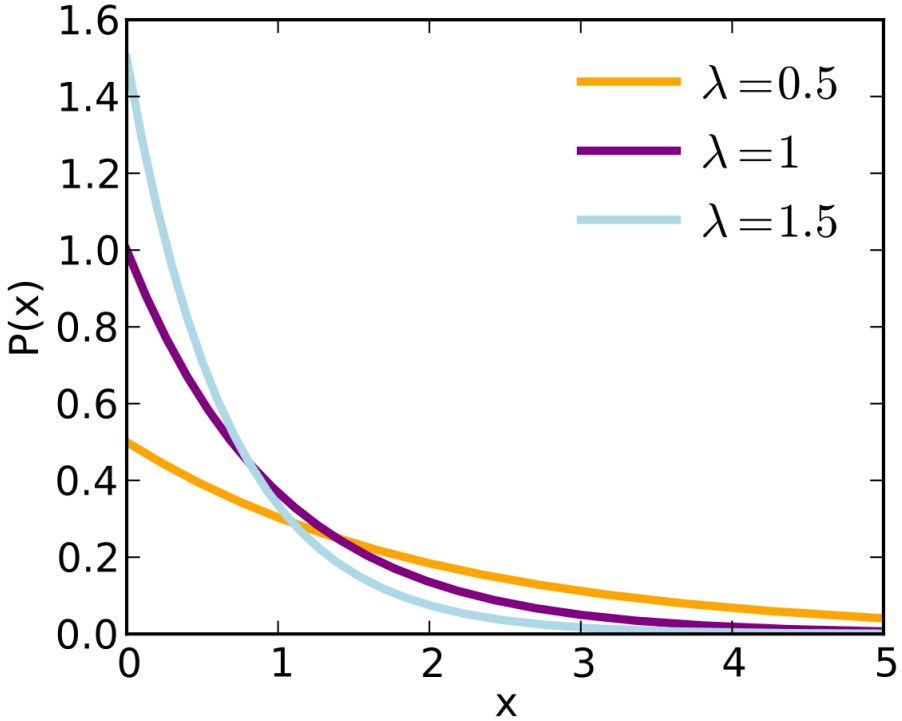
- Multinomial distribution (binomial generalization)
  - Given N observations, it describes the probability to observes m_k times the state in which $x_k = 1$ from a Moultinoulli distribution with mean $\boldsymbol{\mu}$

$$\mathrm{Mult}(m_1, m_2, \dots, m_K | \boldsymbol{\mu}, N) = \binom{N}{m_1 m_2 \dots m_K} \prod_{k=1}^{K} \mu_k^{m_k}$$

Principal probability distributions

- Exponential distribution
  - Useful when we want to concatenate the probabilities close to 0

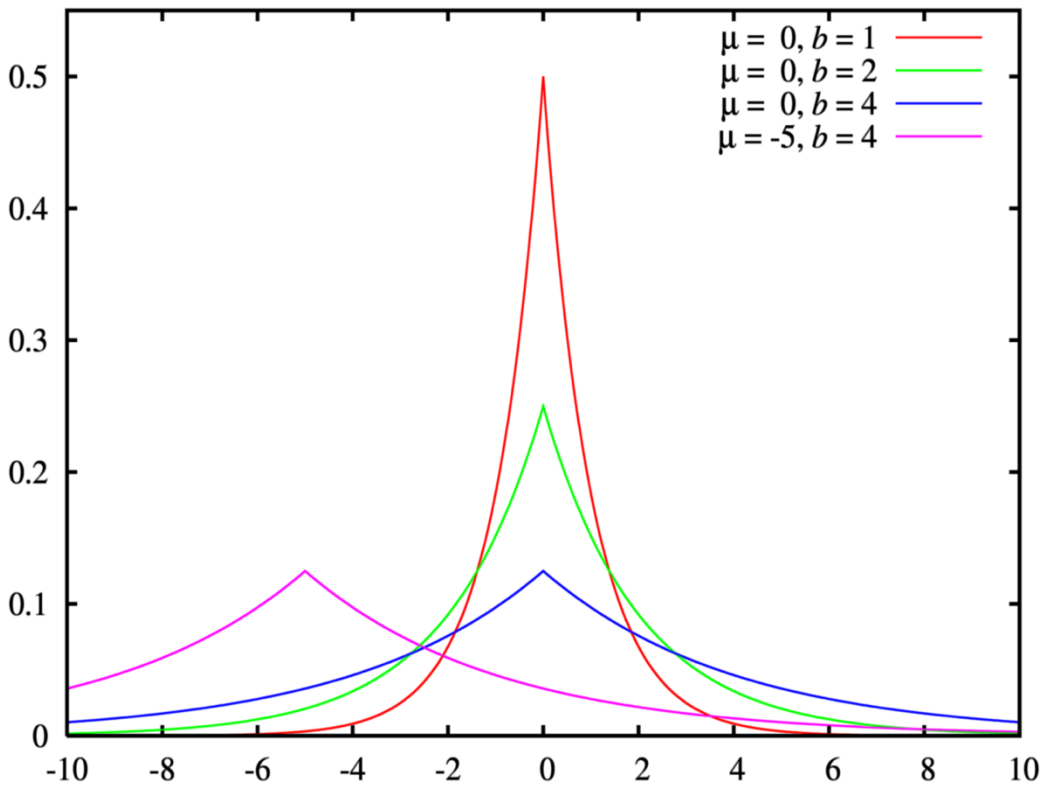
$$p(\mathrm{x};\lambda) = \lambda \mathbf{1}_{x\geq 0}\exp(-\lambda x)$$



Principal probability distributions

- Laplacian distribution
  - Useful when we want to concatenate the probabilities close a given value

$$\mathrm{Laplace}(x;\mu,\gamma) = \frac{1}{2\gamma}\exp\left(-\frac{|x-\mu|}{\gamma}\right)$$



Principal probability distributions

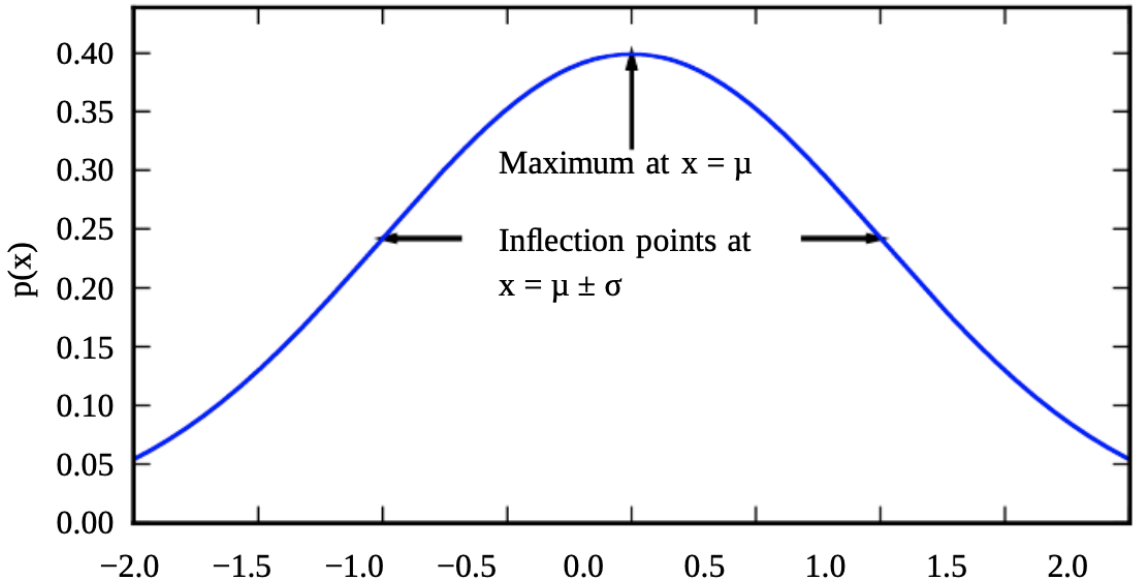
- Empirical distribution
  - Distribution in which a continuous variable is described only by a set of probabilities in only certain points of the domain

$$\hat{p}(\boldsymbol{x}) = \frac{1}{m}\sum_{i=1}^{m}\delta(\boldsymbol{x} - \boldsymbol{x}^{(i)})$$

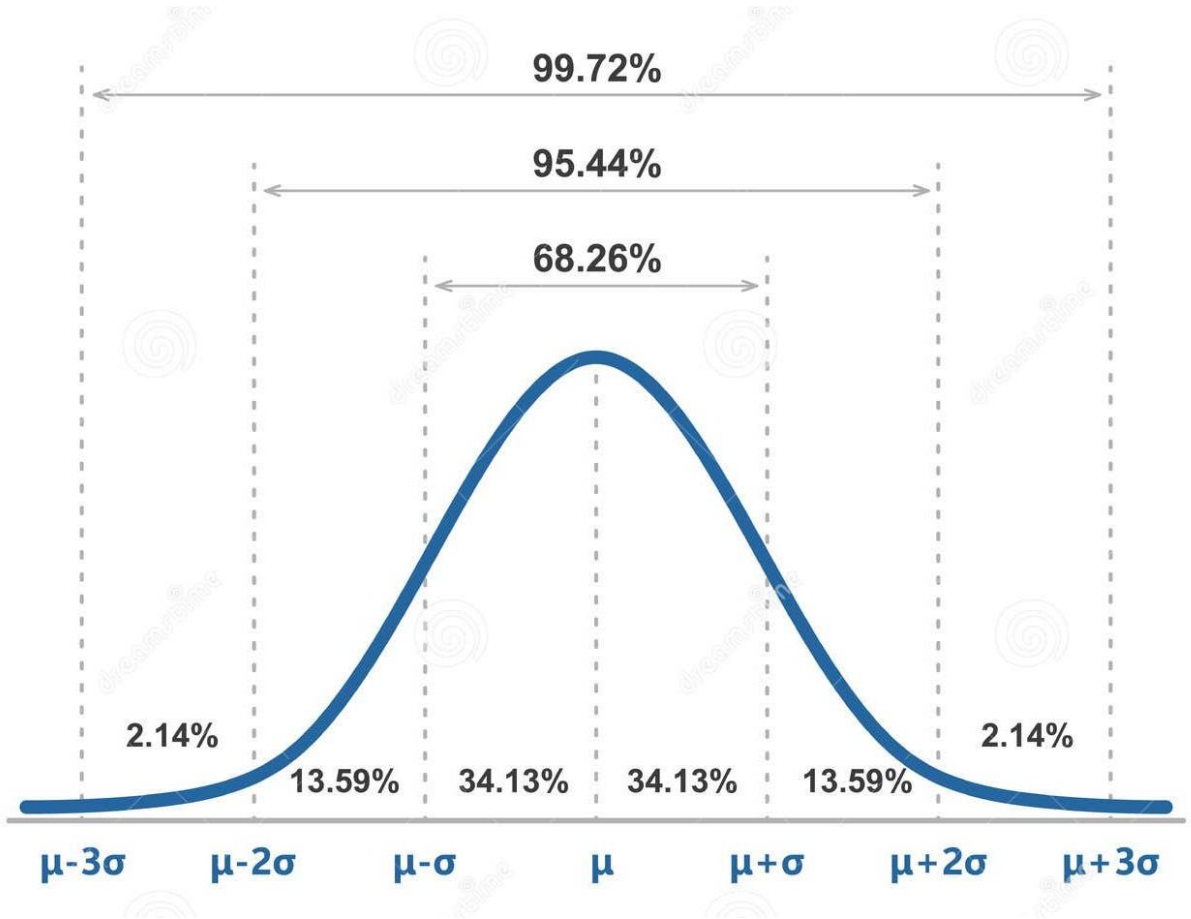
Principal probability distributions

- Gaussian distribution (normal distribution)

$$\mathcal{N}(x;\mu,\sigma^2) = \sqrt{\frac{1}{2\pi\sigma^2}}\exp\left(-\frac{1}{2\sigma^2}(x-\mu)^2\right)$$



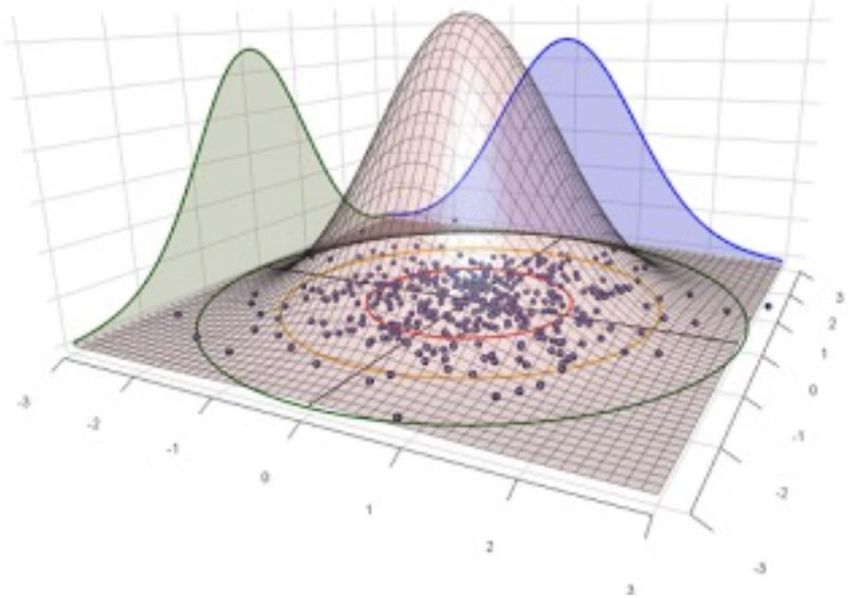
Principal probability distributions



Principal probability distributions

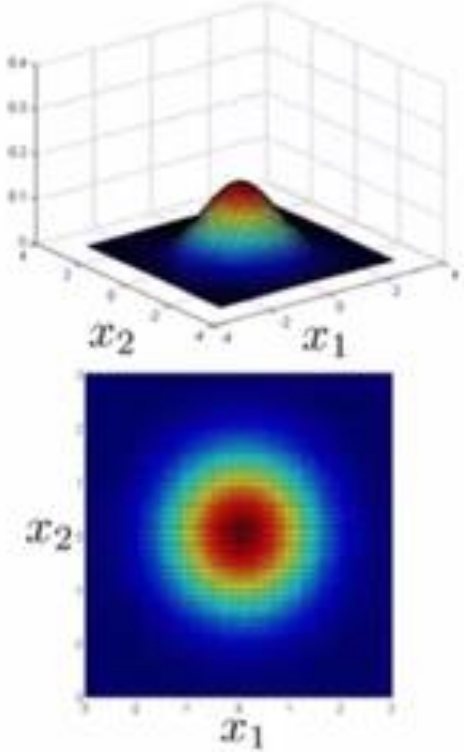
- Multivariate gaussian distribution

$$\mathcal{N}(x;\boldsymbol{\mu},\boldsymbol{\Sigma}) = \sqrt{\frac{1}{(2\pi)^2\det(\boldsymbol{\Sigma})}}\exp\left(-\frac{1}{2}(\mathbf{x}-\boldsymbol{\mu})^T\boldsymbol{\Sigma}^{-1}(\mathbf{x}-\boldsymbol{\mu})\right)$$

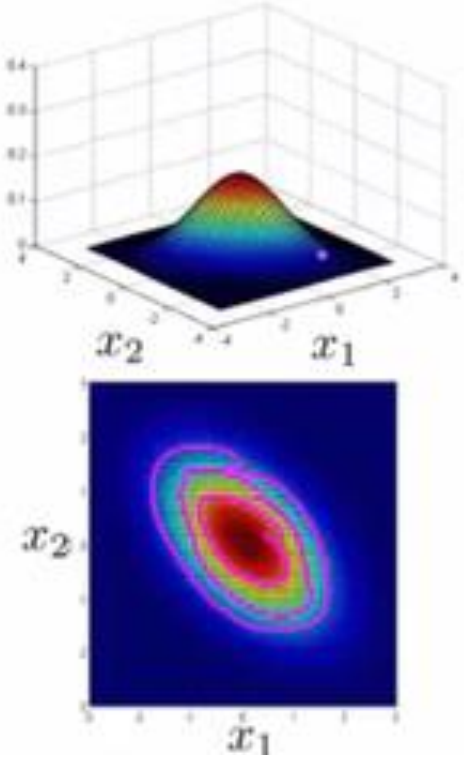


Principal probability distributions

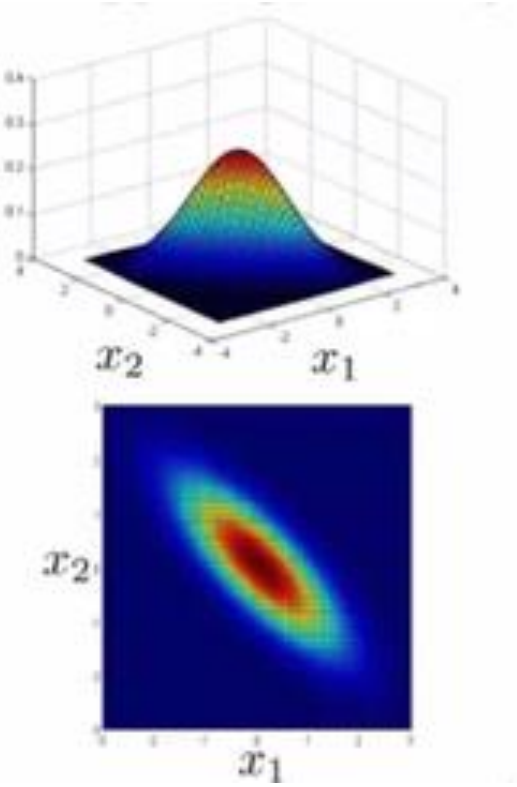
$$\mu = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \Sigma = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$



$$\mu = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \Sigma = \begin{bmatrix} 1 & -0.5 \\ -0.5 & 1 \end{bmatrix}$$



$$\mu = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \Sigma = \begin{bmatrix} 1 & -0.8 \\ -0.8 & 1 \end{bmatrix}$$



Principal probability distributions

- Mixture distribution
  - It represents a complex probability distribution and not known through a set of the component distributions

$$P(\mathrm{x}) = \sum_{i} P(c = i) P(\mathrm{x} | c = i)$$

  - P(c) is a Multinoulli distribution that refers to the i-th component distribution
  - c is a latent variable, a random variable linked to the process that cannot be observed directly
  - Gaussian mixture is a universal approximation of all distributions

Information theory

- The information content of a rare event is higher that a high probable event
- It is possible to sum the information of two independent events

$$I(\mathrm{x} = x) = -\log P(x)$$

$$\log_2 \rightarrow bit\ \ units$$

$$\log_e \rightarrow nat\ \ units$$

Information theory

Shannon Entropy

- It measures the information of an entire distribution as the expected value over it

$$H(\mathrm{x}) = \mathbb{E}_{(\mathrm{x}\sim P)}[I(x)] = -\mathbb{E}_{(\mathrm{x}\sim P)}[\log P(x)] \triangleq H(P)$$

Information theory

Kullback-Leibler divergence

- Measures how are different from each other two distribution *P*(x) and Q(x)
- It is not a distance

$$D_{KL}(P||Q) = \mathbb{E}_{(\mathrm{x}\sim P)}\left[\log\frac{P(x)}{Q(x)}\right] = \mathbb{E}_{(\mathrm{x}\sim P)}[\log P(x) - \log Q(x)]$$

$$D_{KL}(P||Q) \neq D_{KL}(Q||P)$$

$$D_{KL}(P||Q) \geq 0$$

Information theory

Cross-entropy

- It measures the mutual entropy of P and Q
- It has an analogue meaning to D_{KL}

$$H(P,Q) = H(P) + D_{KL}(P||Q) = -\mathbb{E}_{(\mathrm{x}\sim P)}[\log(Q(x))]$$

- It is the typical measure in machine learning, it correlates the real and the predicted distributions

Information theory

- Mutual information
  - Given two random variables X and Y with a correlation, with the distribution P_X and P_Y, MI measures the quantity of information obtained over a variable if the other is observed
  - It is defined as D_{KL} from the joint distribution and the product of the two as in statistical independence

$$MI(X,Y) = D_{KL}\left(P_{X,Y} \parallel P_X \otimes P_Y\right)$$